# Test Plan Document

## Learning management system

**Project Code:**

Group 8-LMS

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**Submission Date:**

_____

**Project Manager's Signature**

# Document Information

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# Definition of Terms, Acronyms and Abbreviations

This section should provide the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

| Term | Description |
|---|---|
| SDS | Software Design Specification |
| UOS | University of Sargodha |
| LMS | Learning management system |
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# 1. Introduction

This document provides a brief and simple introduction to the learning management system(LMS) to be tested. This system has been developed to assist teachers, learners and administrators to easily manage online learning. This system provides an interface to manage courses, upload files, submit assignments, view grades, mark student attendance and communicate with each other.This introduction will assist readers to easily understand what system they will be using and which functionality they will be testing.

## 1.1 Purpose of Document

The purpose of this document is to describe how the learning management system (LMS) will be tested and ensure that all aspect of the system are working properly.This document is helpful in ensuring that the system is functioning as intended and that it meets the needs of student, teacher and administrators.

This document is intended for people who are involved in the development or use of system.This includes people who are testing the system, people who are developing the system, project managers, teachers and administrators.

## 1.2 Project Overview

The project involves designing a learning management system(LMS) for online learning. The system will used by teachers to create courses, upload notes, assign tasks and review student's work. The system will also be used by students to view notes, submit tasks and view grades. The system will also be used by admin personnel to control users and courses.
The primary objective of this system is to ensure that online learning is easy and organized.

# 2. Scope of Testing

*In scope;*

- *User Login*
- *User enrollment in Courses*
- *Study files*
- *Assignments*
- *Marks*
- *Attendences*
- *Messsages*

*Out of scope:*

- *Pay or fee management features*
- *User profile customization*
- *Security*
- *Third party integration*

# 3. Test Plan Strategy

The aim of this test plan is to verify The Learning management system for all functional as well as non-functional requirement specified in SRS.

## 3.1 Unit Testing

**Definition:**

Unit testing is testing each small piece of the learning management system (LMS) individually to ensure it is working properly. Unit testing in this system will test aspect such as login, courses, assignments, file upload, grading and messaging. Each piece is tested individually to identify and correct issues.

**Participants:**

People taking parts in unit testing of the LMS are:

- Developer who build the system
- Test team(if any)
- Other staff helping with testing

**Methodology:**

- The developer will first test their own parts of the LMS.
- Once the module are complete, the test team will test them again to ensure that they are working properly.

## 3.2 Integration Testing

**Definition:**

Integration testing in LMS is checking that all parts like login, courses, study files, assignments, marks, attendance and messages work properly together after being tested individually.

**Participants:**

People taking parts in integration testing of the LMS are:

- Developer who build the system
- Test Team(if Any)
- Other staff helping with testing

**Methodology:**

- Developers combine small components with into larger modules.
- Test team tests these combined modules to ensure they work well together.

## 3.3 System Testing

**Definition:**

System testing checks the whole LMS to make sure all features work together correctly.

**Participants:**

People taking parts in integration testing of the LMS are:

- Developer who build the system
- Test Team(if Any)
- Other staff helping with testing

**Methodology:**

- Test team run the full system with all features
- They check login, courses, assignments, grades, files, attenences and messages.

# 4. Test Environment

Hardware: PC, GPS device
OS: Windows/Android
Software: Web browser(Chrome)
Database: MySQL
Network: Internet connection

# 5. Schedule

| Testing Activities | Begin | End |
|---|---|---|
| Designing Test Cases | 11-NOV-2025 | 17-NOV-2025 |
| Executing Test Cases | 21-NOV-2025 | 25-NOV-2025 |
| Unit Testing | 5-DEC-2025 | 15-DEC-2025 |
| Integration Testing | 5-DEC-2025 | 15-DEC-2025 |
| System Testing | 5-DEC-2025 | 15-DEC-2025 |

# 6. Control Activities

Regular review meetings will be conducted during the testing phase.
The purpose of these meetings is to monitor progress, identify issues, and ensure quality.

**Participants:**

- Project Manager
- Developers
- Testing Team

**Activities:**

- Review test results
- Discuss bugs and errors
- Track testing progress
- Plan next testing steps

# 7. Functions to be Tested

This section lists all the functions to be tested. The following scheme is recommended for listing and description of requirements to be tested.

***7.1. User login***
***7.2. User enrollment in course***
***7.3. Study files***
***7.4. Assignments***
***7.5. Marks***
***7.6. Attendences***
***7.7. Messages***

# 8. Functions not to be Tested

This section lists all the functions not to be tested, along with the reason of not testing, for example," low risk, code reused and previously tested with 100% satisfaction". The following scheme is recommended for listing and description of features not to be tested.

***. Pay or fee management features***

***User profile customization***
***Security***
***Third party integration***

# 9. Test Case Design and Description

*<Summarize the common attributes of test cases. This may include input constraints that must be true for every input in the set of associated test cases, any shared environmental needs, any shared special procedural requirements, and any shared case dependencies. The following scheme is recommended for describing test cases in detail.*

***12.1. Test Case 1***
***12.2. Test Case 2***
**.**
**.**
***12.n. Test Case n***

< ***Example for a test case template*** >

| | | | |
|---|---|---|---|
| **Test Case ID:** | LMS-TC-01 | **QA Test Engineer** | *Tuba* |
| **Test case Version:** | **Version:** 1.0 | **Reviewed By** | Emaan Fatima |
| **Test Date:** | *Date* | **Use Case Reference(s)** | UC-Submit Login |
| **Revision History** | New test case (no previous version) | | |
| **Objective** | To verify that a registered user can log in successfully using valid credentials. | | |
| **Product/Ver/Module** | Learning Management System / Version 1.0 / Login Module | | |
| **Environment:** | 1. Hardware: Desktop / Laptop<br>2. Software: Web Browser (Google Chrome), Serve<br>3. Network: Active Internet Connection | | |
| **Assumptions:** | User is already registered<br>System is running properly | | |
| **Pre-Requisite:** | Valid username and password must exist | | |

| Step No. | Execution description | Procedure result |
|---|---|---|
| 1<br>2<br><br>3<br>4 | Open LMS login page<br>Enter valid username<br>Enter Valid Passoword<br>Click login button | Login page is displayed<br>Usename Accepted<br>Password accepted<br>User logged in successfully |

**Comments**

System should redirect user to dashboard after successful login

☐ Passed ☐ Failed ☐ Not Executed

| | | | |
|---|---|---|---|
| **Test Case ID:** | LMS-TC-01 | **QA Test Engineer** | *Tuba* |
| **Test case Version:** | **Version:** 1.0 | **Reviewed By** | Emaan Fatima |
| **Test Date:** | *Date* | **Use Case Reference(s)** | UC-Submit Assignment |

| | |
|---|---|
| **Revision History** | New test case (no previous version) |
| **Objective** | To verify that a student can successfully submit an assignment. |
| **Product/Ver/Module** | Learning Management System / Version 1.0 / Assigment Module |
| **Environment:** | 4. Hardware: Desktop / Laptop<br>5. Software: Web Browser (Google Chrome), Serve<br>6. Network: Active Internet Connection |
| **Assumptions:** | Student is registered in the system<br>Assignment is already uploaded by teacher |
| **Pre-Requisite:** | Student must be logged in<br>Assignment must be available |

| Step No. | Execution description | Procedure result |
|---|---|---|
| 1<br>2<br>3<br>4<br>5 | Open the LMS and log in with valid username and password.<br>Navigate to the assignment section.<br>Select the required assignment.<br>Upload the assignment file.<br>Click on the submit button. | Dashboard is displayed after login<br>Assignment list is shown<br>Assignment details open successfully<br>File is uploaded successfully<br>Assignment is submitted and confirmation message is displayed |

**Comments**

System should show confirmation message after submission

| | |
|---|---|
| | ☐ Passed ☐ Failed ☐ Not Executed |

**Test Case ID:** LMS-TC-03 **QA Test Engineer** Tuba
**Test case Version:** Version: 1.0 **Reviewed By** Emaan Fatima
**Test Date:** Date **Use Case Reference(s)** UC-Enroll Course

**Revision History**
New test case (no previous version)

**Objective**
To verify that a user can enroll in a course successfully.

**Product/Ver/Module**
Learning Management System / Version 1.0 / Course Enrollment Module

**Environment:**
1. Hardware: Desktop / Laptop
2. Software: Web Browser (Google Chrome), Server
3. Network: Active Internet Connection

**Assumptions:**
User is already registered
Courses are available in the system

**Pre-Requisite:**
User must be logged in

| Step No. | Execution description | Procedure result |
|---|---|---|
| 1 | Open LMS and login | Dashboard is displayed |

2
Go to course section
Course list is displayed

3
Select a course
Course details are shown

4
Click enroll button
User enrolled successfully

**Comments**
System should confirm enrollment and show the course in dashboard

☑ Passed ☐ Failed ☐ Not Executed

**Test Case ID:** LMS-TC-04 **QA Test Engineer** Tuba
**Test case Version:** Version: 1.0 **Reviewed By** Emaan Fatima
**Test Date:** Date **Use Case Reference(s)** UC-View Marks

**Revision History**
New test case (no previous version)

**Objective**
To verify that a student can view marks successfully.

**Product/Ver/Module**
Learning Management System / Version 1.0 / Marks Module

**Environment:**

1. Hardware: Desktop / Laptop
2. Software: Web Browser (Google Chrome), Server
3. Network: Active Internet Connection

**Assumptions:**
Student is registered
Marks are already uploaded by teacher

**Pre-Requisite**
Student must be logged in

## Step No. Execution description Procedure result

1
Open LMS and login
Dashboard is displayed

2
Go to marks section
Marks page is opened

3
Select course/subject
Marks are displayed

**Comments**
System should display correct marks for each subject

☑ Passed ☐ Failed ☐ Not Executed

**Test Case ID:** LMS-TC-05 **QA Test Engineer** Tuba
**Test case Version:** Version: 1.0 **Reviewed By** Emaan Fatima
**Test Date:** Date **Use Case Reference(s)** UC-Check Attendance

**Revision History**
New test case (no previous version)

**Objective**
To verify that a student can view attendance record successfully.

**Product/Ver/Module**
Learning Management System / Version 1.0 / Attendance Module

**Environment:**

1. Hardware: Desktop / Laptop
2. Software: Web Browser (Google Chrome), Server
3. Network: Active Internet Connection

**Assumptions:**
Student is registered
Attendance data is already recorded in the system

**Pre-Requisite:**
Student must be logged in

| Step No. | Execution description | Procedure result |
|---|---|---|
| 1 | Open LMS and login | Dashboard is displayed |
| 2 | Go to attendance section | Attendance page is opened |
| 3 | Select course/subject | Attendance record is displayed |

**Comments**
System should display correct attendance for the selected course

☑ Passed ☐ Failed ☐ Not Executed

# 10. Traceability Matrix

| Sr. No. | Requirement ID | Use Case ID | GUI ID | Test Case ID | Status | Date Completed |
|---|---|---|---|---|---|---|
| **1** | Login | UC-Login | GUI-Login | LMS-TC-01 | Complete | 01-12-2025 |
| **2** | Assignment | UC-Submit Assignment | GUI-Assignment | LMS-TC-02 | Complete | 03-12-2025 |
| **3** | Course Enrollment | UC-Enroll Course | GUI-Course | LMS-TC-03 | Complete | 05-12-2025 |
| **4** | Marks | UC-View Marks | GUI-Marks | LMS-TC-04 | Complete | 08-12-2025 |
| **5** | Attendance | UC-Check Attendance | GUI-Attendance | LMS-TC-05 | Complete | 10-12-2025 |

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# 11. Major Deliverables

*This section lists what will be delivered at the end of the test phase, such as:*

- *Test plan*
- *Test cases*
- *Test summary report*

# 12. Risks and Assumptions

## *Assumptions*

*a) Internet connection will be available*

*b) GPS data will be accurate*

*c) Test environment is stable*

## *Risks*

*d) GPS signal lost may effect tracking accuracy*

*e) Third party API dependency*

# 13. Exit Criteria

*a) All critical test case passed*
*b) No high priority defect remaining*
*c) System meets all functional requirements*

# 14. References

| Ref. No. | Document Title | Date of Release/ Publication | Document Source |
|---|---|---|---|
| **LMS-SRS-01** | Software Requirements Specification (LMS) | Oct 2025 | Project Repository (GitHub / Folder) |
| **LMS-SDS-01** | Software Design Specification (LMS) | Nov 2025 | Project Documentation |

| | | | |
|---|---|---|---|
| **LMS-TP-01** | Test Plan Document (LMS) | Dec 2025 | Project File |

# 15. Appendices

*This section should include supporting detail that would be too distracting to include in the main body of the document.*